

Research Questions

Questions formed only from Chapter 7 extracted wording; answers are exact extracted statements.

ID	Question	Extracted answer	Page
RQ1	Which situation is called priority inversion?	A priority inversion is said to occur in the interval $[t_3, t_6]$, since the highest-priority task τ_1 waits for the execution of lower-priority tasks (τ_2 and τ_3).	4
RQ2	Which protocols are presented in the chapter?	The rest of this chapter presents the following resource access protocols: Non-Preemptive Protocol (NPP); Highest Locker Priority (HLP), also called Immediate Priority Ceiling (IPC); Priority Inheritance Protocol (PIP); Priority Ceiling Protocol (PCP); Stack Resource Policy (SRP).	5
RQ3	How does NPP avoid unbounded priority inversion?	A simple solution that avoids the unbounded priority inversion problem is to disallow preemption during the execution of any critical section.	6
RQ4	How does HLP improve NPP?	The Highest Locker Priority (HLP) protocol improves NPP by raising the priority of a task that enters a resource R_k to the highest priority among the tasks sharing that resource.	8
RQ5	How does PIP avoid unbounded priority inversion?	The Priority Inheritance Protocol (PIP), proposed by Sha, Rajkumar and Lehoczky [SRL90], avoids unbounded priority inversion by modifying the priority of those tasks that cause blocking.	10
RQ6	What problems are not prevented by PIP?	Although the Priority Inheritance Protocol bounds the priority inversion phenomenon, the blocking duration for a task can still be substantial because a chain of blocking can be formed. Another problem is that the protocol does not prevent deadlocks.	21
RQ7	What is the purpose of PCP?	The Priority Ceiling Protocol (PCP) was introduced by Sha, Rajkumar, and Lehoczky [SRL90] to bound the priority inversion phenomenon and prevent the formation of deadlocks and chained blocking.	22
RQ8	What is the maximum blocking result under PCP?	Under the Priority Ceiling Protocol, a task τ_i can be blocked for at most the duration of one critical section.	27